

Generation Interconnection System Impact Study Report

For Queue Project Q11-0052

Olivia Park 138 kV
December 31, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0052. The Interconnection Customer proposes a 213.6 MW generating facility interconnecting at the Olivia Park 138 kV substation in the West Virginia area. The request was received on June 25, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 46 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,186,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Rio Dell to St. John 138kV L is overloaded to 102.3% of its 304 MVA emergency rating for the loss of the American to Urban 345kV. The project contributes 6.4% DFAX (13.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rio Dell to St. John 138kV L	Reconductor line	\$2,186,000	100% (\$2,186,000)
	Total allocated to project		\$2,186,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.